

		$\theta_s = \frac{s}{r} \Rightarrow s = r \theta_s = r \theta_{\min} = 200,000 \times 1.57 \times 10^{-6} = \underline{0.31 \text{ m}}$
18	C	Using $d \sin \theta = n\lambda$ $(1/3.0 \times 10^5) \sin 90^\circ = n_{\max} (600 \times 10^{-9})$ $n_{\max} = 5.56 = 5$ (round down) Hence highest order = 5^{th}; $\therefore$ total number of diffracted images = $5 + 5 = \underline{10}$
19	B	Effective resistance of B and D = effective resistance of C and E